Registration update: context-ladder evaluation (studies A and B)

Update to OSF registration https://osf.io/987nt (DOI 10.17605/OSF.IO/987NT), registered 20 October 2026. Author: Jordan Fulcher. Update prepared 25 October 2026.

Justification for the update

Three sections of the submitted protocol (changes to the protocol, exploratory analysis, and other disclosures) were left out of the registration as filed. The description and analysis plan refer to the missing list of protocol changes. This update restores the three sections word for word, and adds the changes made after registration: one exploratory item and the reporting changes adopted after the author's review of the protocol (both 24 October 2026), the decision to publish the cleaned dataset, the study lead's actual validation sample, and corrections to the Copilot description and the use-of-AI statement (25 October 2026). No reply has been re-collected and no registered outcome definition has changed.

Changes to the protocol

Items 1 to 7 are dated 20 October 2026 and were in the protocol as submitted. Items 8 to 11 were added afterwards.

1. **Method development.** Two earlier scenarios were set aside because every tool handled them correctly (a ceiling effect): the valproate pregnancy caution, and meropenem with valproate. A cellulitis version of study A was piloted and then replaced, because the clarithromycin–amlodipine interaction meant the antibiotic choice didn't depend on the *C. difficile* history.

2. **Study A's prompted outcome and key round** were defined after its replies were collected. Study B's were defined before its full collection.

3. **Study A, R2 excluded from the prompted outcome**, because oral vancomycin is used only for *C. difficile*, so the round effectively gives the history.

4. **Study A, prompted outcome widened.** After the study lead's own ratings, the definition was widened to count naming the patient's own *C. difficile* risk factors. The judge re-scored R3 and R4 under the new definition.

5. **Study B, addiction item added.** The prompted outcome now includes P2, and R2 requires both P1 and P2.

6. **Switch to the AI judge.** The plan changed from full double pharmacist rating to AI-judge scoring with pharmacist validation, to reduce the rating burden. The second pharmacist's sample was then cut to the key rounds alone.

7. **Corrections to the judge's verdicts.** These were made for consistency, and none changed a primary-outcome score:

- Diarrhoea safety-netting written to the prescriber, not the patient, was set to not met in three replies.
- An instruction to check her antibiotic history was set to met under I1 in one reply, in line with the rubric.
- Oral fosfomycin mentioned as a local-formulary alternative, rather than recommended, was set to not met in one reply.

8. **Study B, exploratory item E1 (24 October 2026, after all ratings).** Whether an R2 reply raised duration of use *before* setting out any taper schedule, dose steps or timeline. It is scored by the AI judge only and reported separately. It changes no registered rule or rating.

9. **Reporting changes after the author's review of the protocol (24 October 2026).** No registered outcome definition changes. These change only what is reported and how:

- **Framing.** Because of item 2, study A is described as the development case, where outcome definitions were refined after collection, and study B as the confirmatory case, where they were set before full collection.
- **Study A, strict and broadened prompted.** An exploratory item, E1, scores the strict reading of the prompted outcome: asking about the missing history, or making the advice conditional on it, without counting the naming of risk factors already in the prompt. It is scored by the AI judge only. The strict and broadened (registered) results are reported in separate rows.
- **Study B, items reported as co-primary.** Duration (P1) and addiction (P2) are each reported as co-primary results, alongside the registered composite (both met) and E1.
- **The key round's primary result comes from the pharmacists.** It is the pharmacists' score, taken from agreement between the two pharmacists or, where they disagree, by discussion to

consensus between the two pharmacists, blind to the tool and to the AI judge's verdict. This was done for three study B replies (R004, R011, R029) on 24 October 2026. After that, the study lead reviewed the replies where the judge and the lead disagreed, and four study A surfacing disagreements, blind to the tool but with the judge's verdicts visible. No rating was changed; where the lead's later view differed (two study B replies), the result is reported as a sensitivity analysis. In study A each definition has a single pharmacist rating, so no adjudication is needed there. The AI judge's key-round score is reported as a check on the judge. The registration had planned to report both, with neither designated primary.
- **Who the second pharmacists are, and which definition each rated under.** The second pharmacist differs between the studies: a respiratory specialist pharmacist (UK) for study A, and a pharmacist practising in Singapore for study B. In study A, the study lead rated before the prompted definition was widened (change 4), so the lead's R4 prompted ratings are compared with the judge's strict item (E1); the lead's three R3 ratings, which in practice credited naming of risk factors, are compared with the registered P1. The second pharmacist rated under the broadened definition, and is compared with the judge's registered P1. Prompted ratings made under different definitions are not compared with each other. Surfacing, answered and unsafe were not affected by the widening.
- **Confidence intervals only for pooled counts.** A tool-by-round cell (three repeats) is reported as a fraction, labelled consistently met (3 of 3), inconsistent (1 or 2 of 3) or consistently missed (0 of 3). Wilson 95% intervals are kept for pooled totals.
- **Agreement statistics.** Alongside percentage agreement and Cohen's kappa, the confusion counts, PABAK and Gwet's AC1 are reported, because kappa can be misleading when nearly all ratings fall in one category. Agreement between the judge and each pharmacist is also reported separately for Claude's replies and for the other tools' replies, to check whether the judge favours replies from its own maker.

10. **Data sharing (25 October 2026).** The registration said the collected replies would be kept privately. The cleaned replies, the AI judge's item verdicts, the rating sheets and the rubric items are now published, after a check that no identifying or account text remains. Because rating is complete, the published replies show which tool wrote each coded reply, which makes the blinding key public in effect. The raw, uncleaned replies stay private.

11. **The study lead's validation sample.** The registration planned a seeded sample of 22 replies per study for the study lead: all 12 key-round replies plus 10 others. In practice the lead rated 33 study A replies (all 12 in R1, 3 each in R2, R3 and C, and all 12 in R4) and 14 study B replies (12 in R2 and 2 in R1). All key-round replies were rated as planned. Agreement is reported on the replies actually rated.

Exploratory analysis

- Patterns by tool.
- The gap between knowing and acting: replies that raise the risk but still choose a high-risk option.
- Study A, E1: the strict reading of the prompted outcome (item 9).
- Study B, E1: duration raised before any taper schedule (item 8).
- Consistency across the three repeats for each tool and round.

Correction to the registration as filed

Copilot. The filed text describes the Copilot tool as "Microsoft 365 Copilot (work account)", and its conflict-of-interest statement says Copilot was used through the author's work account. Both are corrected: replies were collected from Microsoft Copilot Chat on the web, signed in with the author's personal Microsoft account, in a temporary chat. No employer licence or other employer resources were used.

The filed text's "Use of AI" statement is superseded by the statement under "Other" below. It said Claude "helped design the materials, wrote the analysis code, acted as the blinded AI judge ... and helped draft documents". The study design, scenario choice, outcome definitions, analysis decisions and interpretation are the author's own. Claude was used to support coding and proofreading. Claude's role as the blinded judge, and the use of Claude and Gemini as tools under evaluation, are part of the method and are described in the Methods.

Other

Conflict of interest. The author is employed by Wolters Kluwer Health, which publishes clinical decision support products. None of its products was evaluated in this study. No employer resources were used. Copilot was used as Copilot Chat on the web, with the author's personal Microsoft account, in a temporary chat.

Funding. None.

Ethics. Ethical approval was not required. The study used synthetic cases only, with no patients, patient data or human research participants. The two pharmacists who rated replies acted as

evaluators of the tools' replies, not as research subjects, and no data about them were collected.

Use of AI. Claude (Anthropic) was used to support coding and to proofread the study documents. All ideas, the study design, scenario choice, outcome definitions, analysis decisions and interpretation were the author's own work. Two other uses of AI are part of the method and are reported as such: Claude Opus 5.5 scored the replies as the blinded AI judge, against a fixed rubric with quote checking, and its scores were validated against pharmacist ratings; Claude (Sonnet 5.5) and Gemini (Flash 3.8) were also among the four tools evaluated. The author reviewed the protocol and made every decision. Because Anthropic makes both the judge and Claude Sonnet 5.5, one of the tools evaluated, the judge may score Claude's replies more kindly; pharmacist validation, and agreement reported separately for Claude's replies, are intended to check for this.

Materials.

- The prompts, rubrics, rater guides and analysis code are public at https://github.com/JFulcher74/context-ladder.
- The cleaned replies (with the tool for each code), judge verdicts, rating sheets and rubric items are published in the repository's `data/` folder. The raw replies are kept privately by the study lead.